Hannah Meibers, Ph.D. (She/Her/Hers)

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EDUCATION

Ph.D., Immunobiology September 2023
University of Cincinnati and Cincinnati Children's Medical Center, OH

Bachelor of Science, Biology April 2017
University of Cincinnati, OH

WORK EXPERIENCE

Arcus Biosciences, Hayward, CA Oct 2023 – Present

Senior Scientist (Biology) Jan 2026 – Present

- Developed and optimized novel *in vivo* mouse models to support pharmacokinetic/pharmacodynamic (PK/PD) characterization.

Scientist (Biology) Oct 2023 – Jan 2026

- Established an inflammation research program by designing and executing key *in vitro* experiments and experimental workflows.
- Contributed to 5+ oncology and inflammation programs spanning preclinical discovery through Phase III clinical development.
- Built and optimized an in-house humanized PBMC mouse model to evaluate therapeutics currently in clinical trials.
- Led development of psoriasis-relevant *in vivo* models to enable preclinical therapeutic evaluation.
- Authored a first-author manuscript through cross-functional collaboration, synthesizing data and communicating findings for a program outside my primary research area.
- Delivered weekly presentations summarizing preclinical program progress, key data, and strategic implications to cross-functional teams.

University of Cincinnati and Cincinnati Children's Medical Center, OH July 2018 – Oct 2023

Graduate Research Assistant
Department of Immunobiology

- Focused on investigating the crosstalk between the adaptive and innate immune system in cases of T cell mediated autoimmune diseases such as multiple sclerosis and type I diabetes.
- Identified a novel proinflammatory pathway in dendritic cells that is activated during T cell mediated autoimmune inflammation involving a DNA damage response and non-canonical STING pathway.
- Invited to discuss my findings at the Keystone Symposia in Snowbird, UT (2023) and the Annual Immunology Retreat in Cincinnati, OH (2022).
- Optimized assays for detecting innate cytokines *in vivo* in peripheral blood (IL-1β, IL-6, IL-12).
- Identified a detrimental role of IL-1R on intraepithelial cells using an ulcerative colitis mouse model using loss of function and neutralizing antibody approaches, thus proposing bench-to-bedside benefits by obstructing IL-22 and IL-1R signaling.
- Characterized innate and adaptive immune cell infiltration in T cell mediated autoimmune disease models with the use of 14 color flow cytometry, intracellular staining, and histopathology.
- Mentored incoming post-doctoral fellows, graduate students and undergrad SURF students, advancing my leadership and time management skills.

Orange Grove Bio, Cincinnati, OH Jan 2022 – Jan 2023

Senior Associate (Biology)

- Assessed pre-clinical development of novel nanotechnologies and immunotherapies created by established Principal Investigators in the Cell Therapy field.
- Evaluated efficacy, toxicity potential, and clinical expectations of innovative immunotherapies.
- Discussed experimental goals with Principal Investigators to expedite their technology to clinical trials.
- Researched and presented information about the competitive fields surrounding these technologies.

SKILLS AND QUALIFICATIONS

Proficient in:

- Sterile mammalian cell culture, microbiology culturing
- T cell, stem cell, and myeloid cell differentiation techniques
- Intracellular and multi-color flow cytometry (Cytek Aurora, BD Fortessa, Novocyte)
- Fluorescence activated cell sorting (FACS)
- Cell cloning (FLAG-tagging, CRISPR knockdown)
- Western blotting and Immunoprecipitation
- ELISA
- RT-qPCR, PCR
- Calcium flux and chemotaxis assays
- Immunohistochemistry
- Bulk and scRNA-seq analyses
- *In vivo* mouse models: humanized mouse models, multiple sclerosis, ulcerative colitis, psoriasis, solid tumorigenic tumor models, regulatory T cell depletion assays, *C. rodentium* infections
- Next-generation sequencing data analysis (R, CLC Workbench, GitHub, Galaxy)
- Computer/Technical (Adobe Illustrator, Linux, MS Office)

PUBLICATIONS AND PRESENTATIONS

Meibers HE*, Reiner GL*, et al. Pharmacological inhibition of both DGKα and DGKζ is required for optimal T cell activation. *Journal of Pharmacology and Experimental Therapeutics*. Currently undergoing revisions. *Co-First Author

Warrick KA, Vallez CN, **Meibers HE**, Pasare C. Bidirectional Communication Between the Innate and Adaptive Immune Systems. *Annual Reviews of Immunology*. 43(1). 2025. doi.org/10.1146/annurev-immunol-083122-040624.

Saha I, Chawla A, Oliveira AP, Elfers EE, Warrick KA, **Meibers HE**, et al. Alloreactive memory CD4 T cells promote transplant rejection by engaging DCs to induce innate inflammation and CD8 T cell priming. *PNAS*. 121(34). 2024. doi.org/10.1073/pnas.2401658121.

Meibers HE, Warrick KA, et al. Effector memory T cells induce innate inflammation by triggering DNA damage and a non-canonical STING pathway in dendritic cells. *Cell Reports*. 42(10). 2023. doi.org/10.1016/j.celrep.2023.113180.

Overcast GR*, **Meibers HE***, et al. IEC-intrinsic IL-1R signaling holds dual roles in regulating intestinal homeostasis and inflammation. *J Exp Med*. 220(6). 2023. doi.org/10.1084/jem.20212523. *Co-First Author

McDaniel MM, Chawla A, Jain A, **Meibers HE**, et al. Effector Memory CD4(+) T cells induce damaging innate inflammation and autoimmune pathology by engaging CD40 and TNFR on myeloid cells. *Sci Immunol*. 7(67). 2022. doi.org/10.1126/sciimmunol.adk0182.

McDaniel MM, **Meibers HE**, Pasare C. Innate control of adaptive immunity and adaptive instruction of innate immunity: bi-directional flow of information. *Current Opinions in Immunology*. 73:25-33. 2021. doi.org/10.1016/j.coi.2021.07.013.

Meibers HE, Finch G, et al. Sex- and developmental-specific transcriptomic analyses of the Antarctic mite, Alaskozetes antarcticus, reveal transcriptional shifts underlying oribatid mite reproduction. *Polar Biol* 42, 357–370. 2019. doi.org/10.1007/s00300-018-2427-x.

Meibers HE, Mitchell CM, et al. TIGIT blockade enhances the effect of anti-PD-1 against CD155hi expressing tumors in mouse and human models [Abstract]. AACR Immuno-Oncology Conference (AACR-IO), Los Angeles, CA. Feb 2026. **Poster Presentation.**

Meibers HE, Warrick KA, et al. *Effector Memory T cells induce innate inflammation by triggering DNA damage and a non-canonical STING pathway in DCs* [Abstract]. Keystone Symposia, Snowbird, UT. April 2023. **Oral and Poster Presentation.**

Meibers HE, Warrick KA, et al. *Effector Memory T cells induce innate inflammation by triggering DNA damage and a non-canonical STING pathway in DCs* [Abstract]. Midwinter Conference of Immunologists, Pacific Grove, CA. Jan 2023. **Poster Presentation.**

Meibers HE, Saha I, et al. *Effector Memory T cells induce innate inflammation by triggering DNA damage and a non-canonical STING pathway in DCs* [Abstract]. Annual Immunology Retreat, University of Cincinnati and Cincinnati Children's Medical Center. Oct 2022. **Oral Presentation.**

Meibers HE, McDaniel MM, et al. *Effector memory CD4 T cells engage myeloid cells through TNFR and CD40 to induce innate inflammation and autoimmune pathology through recognition receptors* [Abstract]. J Immunol, American Association of Immunologists, Portland, OR. May 2022. **Poster Presentation.**

Meibers HE, Jain A, et al. *The role of DC-intrinsic STING during T cell-mediated inflammation* [Abstract]. Annual Immunology Retreat, University of Cincinnati and Cincinnati Children's Medical Center. Oct 2020. **Poster Presentation.**

Meibers HE, McDaniel M, et al. *Vps33B is a crucial regulator of Type I Interferon response downstream of the cGAS-STING pathway* [Abstract]. J Immunol, American Association of Immunologists, Honolulu, HI. May 2020. **Oral and Poster Presentation.** *Due to Covid-19 pandemic, this conference was canceled.